PATENT APPLICATION

Topologically Protected Recording Device with Sectored Architecture and Multi-Stage Signature Chain

TECHNICAL FIELD

The present invention relates to a recording device for generating tamper-resistant audiovisual data, in particular a recording device having a sectored architecture in which a proprietary first sector and an open second sector are unidirectionally connected by a physical data diode, as well as associated methods and platform systems for authenticity verification.

BACKGROUND OF THE INVENTION

Problem Statement

Due to the rapid advancement of generative artificial intelligence, humans are increasingly unable to distinguish genuine recordings from AI-generated images, videos, or audio content. Deepfake technologies enable the creation of photorealistic content that is virtually indistinguishable from authentic recordings. This poses a fundamental threat to trust in visual and audiovisual media, with severe implications for journalism, war reporting, surveillance, the administration of justice, and societal communication.

Prior Art

Known approaches include:

In-Camera Signing (Sony Camera Authenticity Solution, C2PA): Sony implements an in-camera digital signature based on the C2PA standard, in which a signature is generated and embedded in the image at the time of capture. The keys are stored in a hardware chipset. Additionally, 3D depth information is used for detection of screen re-recordings. However, the architecture is monolithic: a single security domain encompasses all cryptographic functions, without topological separation between the manufacturer domain and the user domain.

Blockchain-Embedded Camera Systems (Columbia University, WO2019236470A1): This system combines a digital camera with a hardware security module (HSM) and blockchain-enabled hardware. Hashes of the audiovisual material are published via smart contracts on a public blockchain. GNSS data and spectral analysis serve for location and time verification. This system, too, is monolithically constructed and does not feature sectorization. Tamper protection is achieved through a physical tamper detection box and GPIO-based detection of unauthorized connections.

Multi-Stage Signature Chains Across Separate Devices (Skydio, US12231575B2): Skydio describes an architecture for Trusted Contextual Content (TCC), in which multiple physically separate devices—for example, a drone and its controller—each generate their own signatures and chain them together. Each device signs the entire data package accumulated up to that point, including prior signatures. However, the signatures are distributed across separate physical devices, not across topologically isolated sectors within

a single recording device. A unidirectional hardware data diode between the signing instances is not provided.

PUF-Based Key Generation in Image Sensors: It is known to integrate Physical Unclonable Functions (PUFs) in image sensors to generate unique cryptographic keys from manufacturing-related variations (US11652649B2, US11316705B1). However, these approaches are limited to single-device architectures and do not combine PUF with a sectored dual-signature architecture.

Data Diodes: Hardware-based unidirectional data transmission systems are known in the field of IT/OT security. They are employed in critical infrastructures for network segregation. Their use within a recording device for topological separation of cryptographic domains is, however, not known.

Further Individual Aspects of the Prior Art: LegitIPix (US10439821B2) describes direct signature embedding in pixel data of a CMOS sensor using a shared-key approach. NXP (US20250202718A1) discloses an eIDAS-based certificate provisioning via a provisioning device. Amazon (US11770260B1) teaches a frame-by-frame hash chain for video verification. Basler (US20210385408A1) discloses an authentication module for sensor data with steganographic modification. EP3262782B1 describes an anti-tamper system with decoupled architecture and zeroization.

What the known solutions have in common is that they feature a monolithic security architecture in which a single trust domain encompasses both the sensing elements and the entire cryptographic processing. This means that the manufacturer or a single attacker with access to this trust domain can potentially compromise the entire system. There is therefore a need for an architecture that substantially increases tamper resistance through topological separation and physically enforced unidirectionality.

SUMMARY OF THE INVENTION

The invention solves the stated problem by means of a recording device having a sectored architecture, comprising:

a) a first sector comprising at least one sensor and a first signature engine, the first signature engine being configured to determine cryptographic hash values from a signal captured by the sensor, and to digitally sign said hash values with a private key of a key pair assigned to the first sector, so as to generate a first signature;

b) a second sector comprising at least a second signature engine, the second signature engine being configured to receive the signal and the first signature and to generate a second signature using a private key of a key pair assigned to the second sector;

c) a data diode disposed between the first sector and the second sector, which effects a physically enforced unidirectional data transmission from the first sector to the second sector and prevents information backflow from the second sector to the first sector;

wherein the first sector and the second sector are assigned to different trust domains.

DETAILED DESCRIPTION

The invention is described below by way of example with reference to a video camera. This example is non-limiting; the concept is transferable to all recording devices, including audio recorders, surveillance cameras, body cameras, and smartphones.

1. Sectored Architecture

1.1 First Sector (Proprietary Sector)

The first sector is attributable to the manufacturer of the recording device and is preferably configured in a manner that is as manufacturer-specific as possible. It is preferably arranged wholly or partially within the housing of the recording device, and for example wholly or partially within the second sector. Within the first sector there are located at least:

- an image sensor for capturing visual data;
- optionally further sensors, in particular microphones for capturing acoustic data;
- a first signature engine which generates cryptographic hash values from the audiovisual signal and signs these with a private key of a key pair assigned to the first sector.

The protection of the first sector is achieved primarily through secrecy and manufacturer-specific design. Preferably, the first signature engine is implemented as a system-on-chip (SoC) directly on the chip of the image sensor, in order to minimize the path between data source and signing point.

The signing is performed close to the source. Preferably, not only the raw data delivered by the image sensor and any microphones, but also metadata generated in the first sector—in particular autofocus information, depth sensor data, liveness parameters, and/or other sensor data of the first sector—are included in the hash value computation and signed therein jointly before leaving the first sector. Preferably, the cryptographic hash values are determined on a per-frame basis.

1.2 Second Sector (Open Sector)

The second sector is attributable to the user of the recording device. In contrast to the first sector, protection is achieved here not through secrecy, but through maximum transparency. Preferably, the second sector is configured as an open-source system whose hardware and software are publicly viewable and verifiable. Preferably, the second sector is likewise arranged wholly or partially within the housing of the recording device.

Within the second sector there is located at least a second signature engine. This receives the audiovisual signal together with the first signature of the first sector via the data diode and generates a second signature.

Preferably, in the second sector, additional metadata are added to the audiovisual signal, which are collected by means of additional sensors, in particular:

- a GNSS module for position determination;
- a long-wave time signal receiver (e.g., DCF77 at 77.5 kHz) for an independent time reference;
- inertial sensors including gyroscopic sensors and accelerometers;
- additional environmental sensors.

1.3 Data Diode

A data diode is disposed between the first and the second sector. The term data diode is to be interpreted broadly and encompasses any device that effects a physically enforced unidirectional data transmission. The data diode may be realized in particular by:

- optoelectronic couplers (optocouplers), in which a light-emitting component in the first sector drives a photosensitive component in the second sector, without a practically usable return channel physically existing (the parasitic capacitance of the optocoupler is to be taken into account by a person skilled in the art, since it opens up a hypothetical possibility of back-coupling);
- fiber-optic connections with exclusively one light-transmitting unit and one light-receiving unit;
- circuits with strongly asymmetric transfer characteristics, in which signal transmission in the reverse direction is physically prevented or attenuated to such an extent that information backflow is not possible.

It is critical that at the interface between the sectors, data can flow exclusively in one direction—from the first to the second sector. Thereby, an attacker who has compromised the second sector is physically precluded from acting upon the first sector across the data diode.

2. Signature Chain

2.1 First Signature (Proprietary Sector)

In the first sector, cryptographic hash values are determined from the audiovisual signal. Preferably, this is done on a per-frame basis, i.e., for each captured individual frame a separate hash value is generated. The hash values are signed with the private key of the first sector, thereby generating the first signature.

It is possible to compress the recording data prior to hash value computation. This may also be done in parallel: in the first sector, a RAW stream may be generated and signed on a per-frame basis, and from the RAW stream, reversibly or irreversibly compressed streams may be generated that are likewise independently signed. The data diode then forwards both the RAW stream and the compressed streams together with their respective signatures to the second sector.

2.2 Second Signature (Open Sector)

In the second sector, the received data package—consisting of the already fully signed data package provided by the first sector (audiovisual data and optionally metadata of the first sector) and the first signature—together with the metadata collected in the second sector is subjected to a second signing. The second signature engine generates cryptographic hash values over the complete data package and signs these with the private key of the second sector.

The dual signature chain ensures that a forgery requires the compromise of both independent sectors. Since these are assigned to different trust domains (manufacturer vs. user) and are physically separated by the data diode, the tamper resistance is substantially increased compared to monolithic systems.

3. Key Generation and Key Management

3.1 PUF-Based Key Generation

In a preferred embodiment, the generation of the private keys involves a Physical Unclonable Function (PUF). A PUF exploits manufacturing-related, unavoidable variations in the semiconductor structure to generate a unique, hardware-bound entropy contribution.

The private key of each sector is preferably composed of at least two components:

- a PUF-derived component, which is obtained from the physical structure of the respective sector;
- a manufacturer-side or user-side component, which is introduced during initialization.

Through this partitioning, neither the manufacturer alone nor the user alone knows the complete private key of their respective sector. Additional "key sources" may be added, with preference given to those where collaboration with the manufacturer or user is implausible. The PUF hardware itself also does not "know" the complete key, since it merely provides an entropy contribution. Preferably, after the PUF hardware has delivered its entropy contribution, it is irreversibly altered, which in systems based on MOS structures can be achieved by applying an otherwise impermissible gate-source voltage V_GS. In this context, a sufficiently dimensioned voltage surge is preferably applied to destroy all gates of the MOS structure such that even electron-microscopic examinations no longer provide any information about a prior memory state.

In this way, it is achieved that, after initialization of the recording device, the private keys exist exclusively in volatile memories, specifically in memories of the camera that are protected against readout, and there in at least two disjoint such memories (namely at least one in the first and at least one in the second sector).

The described method, in which a key is determined or obtained from an external contribution and a contribution of a PUF, and is then stored in a volatile memory, whereupon the PUF is irreversibly altered, is not limited to recording devices, but can be used in all applications in which the loss of a hardware-bound key causes lower costs than a theft of the same.

3.2 Key Storage

The private keys are preferably held in volatile memory (e.g., SRAM) that is permanently powered by a battery. A loss of supply voltage leads immediately to the loss of the keys. In combination with the anti-tamper measures described below, comprehensive protection against physical key extraction is thereby achieved.

4. Tamper Protection (Anti-Tamper)

4.1 Gas Monitoring

In a preferred embodiment, at least the first sector of the recording device is filled with a gas, preferably an inert gas, whose pressure deviates significantly from ambient pressure. Throughout the entire service life of the recording device, the pressure and temperature of the gas are measured quasi-continuously.

A memory zeroization is triggered as soon as one of the following conditions is detected:

- an excessive rate of pressure change, indicating an opening of the housing;

- an absolute pressure that is too high or too low;
- an excessive deviation of the sensor data from the gas law, which may indicate a change in the effective sector volume (opening) or a manipulation of the sensor system itself;
- an impermissibly low temperature, in particular for protection against cold-boot attacks, in which an attacker attempts to preserve and read out the volatile memory at low temperature.

The pressure measurement is preferably temperature-compensated, in order to avoid false alarms (and consequently bricking) caused by temperature changes during normal operation. The described gas monitoring also provides protection against non-thermal data remanence attacks such as, for example, "volt boot" attacks, since it substantially impedes successful physical access to the key memory. Additionally or alternatively, monitoring of the chemical composition of the gas fills is possible, for example with electrochemical or optical sensors, wherein the exact composition of a gas fill from the manufacturer is preferably a manufacturer secret. The described gas monitoring with pressure-dependent memory zeroization is not limited to recording devices, but can be employed in any electronic device that is intended to protect a cryptographic key held in volatile memory against physical access.

4.2 Thermal and Chemical Destruction

Preferably, upon detection of a manipulation, not only the memory contents are zeroized, but additionally a thermal destruction of the memory cells is initiated, for example by short-circuiting a battery arranged in the respective sector for maintaining the memory state. For example, the waste heat of a fuse associated with a battery may be used directly or indirectly (e.g., by igniting a reactive composite such as Nanofoil™) for thermal destruction. The fuse itself may also comprise a reactive composite ("Pyrofuze"). Thereby, the memory cells and their contents are irreversibly destroyed.

Additionally or alternatively, a destruction of the memory by chemical reaction with atmospheric constituents may be provided, which commences upon loss of housing integrity.

4.3 Further Tamper Protection Measures

A combination with further known forms of tamper protection is recommended, including prestressed glasses, sensor wires, conductive mesh structures, and similar means.

5. Metadata and Plausibility Verification

5.1 Long-Wave Time Signal and GNSS Cross-Validation

The combination of a long-wave time signal receiver and a GNSS module within the second sector is particularly advantageous. This enables an immediate plausibility check:

- Comparison of timestamps: The long-wave time signal (e.g., DCF77) provides a time reference independent of satellite navigation. A comparison with the GNSS time allows detection of GNSS spoofing.
- Comparison of the location defined by the GNSS data with the long-wave time signal, taking into account the propagation time from the transmitter to the camera: If the location of the long-wave transmitter is known (e.g., Mainflingen for DCF77), the approximate distance to the transmitter can be inferred from the reception time and the signal propagation time, which must be consistent with the GNSS location.

5.2 Atmospheric Environmental Signals as Fingerprints

The antenna of the long-wave time signal receiver can additionally be used to measure unpredictable natural atmospheric signals, which occur independently of electrical power grids, in particular:

- "radio hiss," VLF hiss (Very Low Frequency noise);
- sferics (electromagnetic pulses from lightning discharges);
- tweeks (dispersive long-range propagation of sferics in the Earth-ionosphere waveguide structure);
- fluctuations in the signal strength of the time signal transmitter.

During a live broadcast to the internet, it would be immediately noticeable if, for example, the VLF hiss of one camera deviates from that of similar cameras in the vicinity.

For verification in particular of sferics, observation infrastructures can be consulted, such as in particular Earth observation satellites, networks of geophysical sensors, or lightning location networks.

5.3 Inertial Sensors for Consistency Verification

Inertial sensors (accelerometers, gyroscopic sensors) in the second sector capture the motion of the recording device. By comparing the camera motion with the image content, plausibility can be verified: if the camera rotates but the image remains stable (or vice versa), this indicates a manipulation.

6. Anti-Screen-Recording (Protection Against Screen Re-Recording)

A further aspect of the invention relates to protection against re-recording of forged images or videos, where even lens attachments with built-in displays would be conceivable. Against this, the following are proposed in particular:

- depth sensors, wherein preferably known or existing autofocus systems are incorporated;
- multiple systems, in particular also those with active object illumination;
- systems that interact with the object to be recorded via ultrasound;
- liveness sensors, in particular for recording of humans.

The image sensor itself may serve as a liveness sensor if it is configured as a 4-channel sensor (RGB-IR sensor). Otherwise, an additional infrared-sensitive image sensor may be provided. Their data are not only used for color correction but are treated as part of the recording.

Signals from the above-mentioned sensors are fundamentally accorded the same significance as those from the image sensor and the microphones, and they are likewise signed close to the source. In this context, imaging sensors are preferably assigned to the first sector, while illumination devices comprising a light source are assigned to the second sector.

Since the first sector does not interpret the data contained in the modulated light, but merely captures and signs them as part of the audiovisual signal, the informational isolation of the first sector is preserved; a falsification of the modulated data by a compromised second sector would be detectable upon external verification.

Illumination devices comprising a light source in the second sector may not exclusively serve for anti-screen-recording protection, but may transmit data to an image sensor of the first sector via object

illumination, which data are then signed for the first time in the first sector as part of the recording. A modulatable light source of the second sector may, for example, serve both as part of a known active autofocus system and, for instance amplitude- or direction-modulated, carry data useful for subsequent authentication of the recording, which become "visible" in the recording (optionally also in the infrared range, in particular NIR). These data may comprise, for example, unencrypted or encrypted (with a private key of the second sector) geolocation data including GNSS time, time obtained by other means, identity of the camera, and an environmental fingerprint (in particular a radio fingerprint, and therein preferably a VLF fingerprint).

The latter concept of data-carrying object illumination can be extended: Independently of the present invention according to claim 1, an illumination device for object illumination can be presented which, as an accessory to existing cameras with C2PA support, improves trust in their recordings by illuminating objects to be recorded in a correspondingly "data-carrying" manner; the recordings may then contain, for example, a C2PA-signed environmental fingerprint, preferably a VLF and/or shortwave fingerprint.

7. Blockchain Registration and Platform Integration

7.1 Publication of Public Keys

The manufacturer publishes the public key of the first sector in a practically unforgeable manner, in particular in a blockchain, wherein metadata of the camera, in particular a digital fingerprint of its first sector, may be associated with the public key. This fingerprint may comprise a mapping of the pixel defects of the image sensor. Additionally or instead of a pixel defect mapping, the fingerprint may comprise a signature derived from the Photo-Response Non-Uniformity (PRNU) of the image sensor (the PRNU describes the manufacturing-related, unique for each image sensor, spatial pattern of slight sensitivity differences of individual pixels, which is statistically extractable from a plurality of images, and which enables a forensic identification of the image sensor even if the cryptographic key has been compromised.)

The forensic value of such fingerprints, uniquely assigned to an image sensor and published in a practically unforgeable manner, would be immense.

As regards the user, the user is enabled to likewise publish the public key of the second sector (comprising the open-source signature engine) in the blockchain, and to unambiguously cryptographically link it with the public key of the manufacturer.

7.2 Authenticity Verification

For authenticity verification, a cryptographic hash value is newly computed from the audiovisual material at hand, including metadata. The first signature is then verified by means of the public key of the first sector deposited in the blockchain, and the second signature is verified by means of the deposited public key of the second sector. The successful verification of both signatures confirms both the integrity and the authenticity of the recording.

7.3 Platform and Browser Integration

In a further embodiment of the invention, video platforms, social media platforms, web browsers, or browser applications are proposed which, for each uploaded purported recording (audio, video, image),

verify whether it can be attributed through an unbroken certificate chain to a recording made with a recording device that performs close-to-source digital signing. The platform signals this to the user in a comprehensible manner or enables the user to preferably exclusively view recordings verified as authentic.

8. Further Notes on Protection Against Attacks

8.1 Cross-Sector Key Zeroization

Preferably, both the first and the second sector are quasi-continuously tamper-protected such that, in the case of all presumable manipulation attempts, an immediate zeroization of the respective private keys is achieved. Further preferably, a triggering of a tamper protection device in one sector even causes a zeroization of all private keys in all camera sectors, however without digital information crossing sector boundaries for this purpose. This can be achieved, for example, thermally or mechanically, in that a fuse of a battery that powers the volatile key memory in one sector inevitably co-triggers one or more corresponding fuses assigned to other sectors.

For this purpose, a simply connected Al/Pd wire ("Pyrofuze") may be used, in which a first portion serves as an active fusible wire in the first sector and a second portion serves in the second and optionally further sectors; in this way, the command for key zeroization can cross a sector boundary as a diffusion-controlled chemical reaction, without an electrical or electromagnetic signal being transmitted. The described cross-sector key destruction, in which by means of a reactive element an exothermic reaction front propagates across a sector boundary and thereby effects the zeroization or destruction of key memories in multiple security domains, without an electrical or electromagnetic signal crossing the sector boundary, is not limited to recording devices, but can be employed in any electronic device that has two or more physically separated security domains with respective dedicated key memories.

8.2 Protection of the Data Diode

According to the invention, the only information-carrying connection between the sectors consists of one or more data diodes. Thus, the inputs and outputs of the data diodes are located in different sectors. Preferably, at least one tamper protection device of one of the sectors is configured to also detect manipulation attempts that affect the transmission path of the data diode, which may lie outside both sectors.

8.3 Signal Paths as Means of Manipulation Detection

In recording systems according to the invention, the image sensor is arranged within the tamper-protected first sector, which itself may lie within the second sector, and can be optically stimulated externally, while physical interference is impeded. Likewise, the signal path or paths between the image sensor and the signature engine(s) lie in a tamper-protected manner within the first sector. These signal paths can be monitored and may thus themselves form part of a manipulation detection device.

8.4 Inductive and Capacitive Attacks

For protection against electromagnetic attacks, it is proposed to electromagnetically shield the sectors, and particularly the first sector, in a manner known to a person skilled in the art, with appropriate filters for technically unavoidable feedthroughs. Furthermore, by coating, for example, the rear lens element or a

protective glass of the camera on one or both sides with an electrically conductive layer (an ITO layer can additionally serve for reflection reduction), the first sector can form a nearly hermetic Faraday cage. The term glass or protective glass is to be interpreted broadly and shall herein include optical glass-ceramics. Preferably, such electrically conductive layers are galvanically connected to an electrically conductive housing of the first sector.

If gas monitoring is employed, the lens or protective glass may be connected to the sector housing via a permanent seal and must be dimensioned to permanently withstand the pressure differential between the sector pressure and the atmospheric pressure. A material-bonded connection between the "glass" and the camera housing is preferred, which is made possible by means of active solders or by partial metallization of the glass. In this context, the different coefficients of thermal expansion between the optical materials and the housing material are to be taken into account by a person skilled in the art, for example by mounting the "glass" in Invar or Kovar and soldering it thereto, and with due consideration of possible mechanical stresses, the Invar or Kovar element is in turn joined to the camera housing by a material bond (by soldering or welding).

Furthermore, to avoid common-mode interference—including deliberate interference—within the sectors, a consistently differential signal transmission may be employed, at least outside of integrated circuits. Finally, attack-vulnerable electronics may be potted, wherein the potting compound may be selected to be insoluble in organic solvents including halogenated solvents, and may contain fillers to influence its thermal conductivity (e.g., AlN) or its electromagnetic properties (e.g., ferrite powder). Further preferably, the potting compound is selected such that all solvents, including supercritical fluids, that can attack, substantially swell, or even dissolve it, can also attack and destroy the potted electronics. In addition to such passive protection, active protection is also possible here: as soon as an electromagnetic attack is detected, for example by an overvoltage event, key zeroization is triggered immediately.

8.5 Tamper Protection by Missing Response

Furthermore, at least in the second sector, a key zeroization can be triggered as soon as an expected signal from the data diode fails to arrive, since this may be indicative of an attack directed against the first sector. The expected signals from the data diode need not follow a regular interval, but preferably follow one another at pseudo-random intervals.

8.6 Permanent Cryptographic Self-Observation

In a further preferred embodiment, the recording device is configured to generate, throughout its entire operating life—including outside of regular recording operation—at preferably pseudo-random time intervals whose expected value preferably lies in the range of seconds to a few minutes, compressed background images of low resolution by means of the image sensor of the first sector. Each background image traverses the regular signature path: it is signed close to the source in the first sector, unidirectionally communicated to the second sector via the data diode, and there subjected to a second signing together with the current metadata—in particular GNSS position, long-wave time signal, radio environmental fingerprint, and inertial sensor data. Additionally, each chain element thus generated is cryptographically chained with the preceding one by including the cryptographic hash value of the preceding chain element in the subsequent one, so that a seamless, tamper-evident image chain is created over the entire service life of the device. The image chain is stored locally in the second sector in a symmetrically encrypted manner, wherein the symmetric key is known exclusively to the user. If the

authenticity of a recording made with the recording device is disputed, the user can decrypt the entire image chain or—by means of a Merkle tree formed over the chain elements—a verifiable temporal section thereof and present it for review; the integrity of the chain is verifiable by seamless recomputation of the hash values, and any subsequent manipulation, insertion, or deletion of an image destroys the consistency of the chain. By keeping the image resolution sufficiently low and applying sufficiently strong data compression, such a camera can document its integrity on commercially available mass storage devices over many years. Object illumination, e.g., NIR object illumination, as per Section 6, can be incorporated into the protection concept.

9. PQC

Based on the strategic conjecture that NP is not a subset of BQP, in anticipation of powerful quantum computers, various new encryption and signature algorithms are being developed, so-called PQC algorithms. Preferably, at least one signature engine of the first or second sector employs a PQC algorithm. Further preferably, both the first and the second sector use signature engines with PQC algorithms, and preferably different ones, and further preferably ones that are based on different mathematical problems which cannot be efficiently reduced to one another, in particular selected from: lattice-based problems, code-based problems, hash-based schemes, and/or multivariate polynomial equation systems.

10. Blockchain

In addition to the camera itself, a compromise of the blockchain is also conceivable, in which the public key of the manufacturer is deposited in a uniquely linked manner with that of the user. For this purpose, it is proposed to deposit these linked key pairs in a plurality of established blockchains, so that an attacker would have to simultaneously overcome the independent consensus mechanisms of a plurality of blockchains.

11. Powering On and Off of the First Sector

Insofar as powering on and off of the first sector is required, this is preferably done from the second sector, which preferably surrounds the first, by switching the supply voltage on and off. It is to be ensured by hardware means that no changes to the firmware in the first sector can thereby be effected.

Preferably, the firmware of the first sector distinguishes between two operating modes based on the duration of the applied supply voltage:

- In the case of a short voltage pulse whose duration is below a predetermined threshold duration, the first sector generates, after startup, a single signed image (challenge-response frame) and outputs it via the data diode to the second sector, whereupon the second sector switches the supply voltage off again.
- In the case of a supply voltage that is applied beyond the threshold duration, the first sector commences regular audiovisual recording operation.

The second sector monitors the response time of the first sector to the voltage pulse. If the expected response fails to arrive within a predetermined time window, or if the response time deviates impermissibly from a time profile characteristic of the first sector, this is evaluated as a manipulation

indicator and may trigger a key zeroization in accordance with Section 8.5. The challenge-response frames generated at pseudo-random intervals are preferably incorporated into the image chain in accordance with Section 8.6. This serves to defend against side-channel attacks.

12. Digital Terminal Devices

The integration of recording devices according to the invention into digital terminal devices such as smartphones enables anyone to make recordings of the highest evidentiary value.

13. Definitions

A "sector" within the meaning of this application designates a physically delimited functional area within a single device that comprises its own processing units, its own memory, and its own cryptographic infrastructure, and that is informationally isolated from other sectors of the same device—that is, between the sectors there exists no intended or unintended information path other than the provided data diode, wherein side channels such as electromagnetic emanation, power consumption patterns, thermal coupling, or shared clock sources are suppressed by suitable design measures.

A "trust domain" is the totality of those hardware and software components whose integrity is under the control of exactly one responsible entity—such as the manufacturer or the user—and that are assigned to a common root of trust; different trust domains are characterized by the fact that none of them need presuppose the integrity of any other.

A root of trust is that component within a trust domain whose authenticity and integrity is not derived from a further entity within the same domain, but is taken as given—typically a cryptographic key or a hardware component housing such a key, from which the chain of trust for all remaining components of the domain originates.

A data diode is a physical connection between two sectors that restricts the information flow by hardware means to exactly one direction and physically prevents a backflow—unlike a merely software-based access control.

A signature engine designates a unit integrated in the respective sector that forms cryptographic hash values from received data and digitally signs these by means of a private key exclusively assigned to the sector.

Close to the source means that the signing is performed spatially and in terms of circuit design so close to the sensor that no unsecured, externally accessible signal path exists between data capture and signing.

14. Internet-Capable Cameras and Live Broadcasting

Recording devices according to the invention may, as already mentioned, be internet-capable and in particular be configured for live broadcasting of audiovisual content. In a preferred embodiment, the recording device is configured to cooperate with an external provenance or time-anchoring protocol which provides for anchoring cryptographic hash values with one or more device-external services. Such external services include in particular Time Stamping Authorities (TSAs) according to RFC 3161, publicly auditable Transparency Logs as provided by the IETF SCITT architecture, and blockchains. Examples of provenance protocols that build on one or more of these anchoring mechanisms are the C2PA

standard in its respective version, the Content Provenance Profile (CPP), and the protocols of blockchain-based timestamping services. The invention is not limited to any particular protocol.

The sectored architecture of the invention and such provenance protocols complement each other advantageously: Known provenance protocols establish an externally verifiable time binding of the recordings, but make no statement about the physical integrity of the recording device itself: a device compromise prior to the recording typically lies outside their threat model. The inventive sectorization with physical data diode ensures precisely this device integrity. The combination of both methods thus provides a hitherto unattained level of security, since both the time of the recording (through the external service) and the authenticity of the recording device (through the dual signature chain) become independently verifiable. This creates trust that the timestamped and signed purported recording actually is a recording and neither a playback nor material generated in quasi-real time.

A provenance protocol core is preferably executed in the second sector (open sector), since this may have the required network connectivity, and the provenance protocol there processes exclusively data already signed in the first sector, received via the data diode (plus the metadata generated in the second sector). An attacker who compromises the provenance process in the second sector cannot forge the first signature of the proprietary sector. Alternatively, the protocol core may be executed outside the second sector, and preferably in a third protected sector, but in no case in the first sector, since the integrity of the latter is to be kept free from network connectivity and the associated attack possibilities.

In particular, embodiments of the invention that record radio signals (sferics, tweeks, (VLF) hiss) or information extracted from radio signals according to Section 5.2 as metadata gain considerably in credibility through an additional external time anchoring: The atmospheric radio signal, for instance, exists unpredictably only at a specific location at a specific time and cannot be predicted; the at least one external service provides an independently verifiable proof of time. Together, a location-time binding arises that no single actor can forge, since the external service cannot know the local radio fingerprint, and a local attacker cannot manipulate the external proof of time.

A single TSA, however, constitutes a single point of failure and a rewarding attack target, the compromise of which may go undetected. And even a Transparency Log is only as trustworthy as its auditors:

To avoid concentrating trust in a single external service, the following improvement is proposed: The recording device preferably connects to a plurality of mutually independent external services, which preferably belong to different categories, for example a TSA and a blockchain, or a Transparency Log and a TSA. The selection of the specific services is preferably made by a transparent random or pseudo-random procedure, so that the selection is retroactively verifiable but not predictable for an attacker. For example, the pseudo-random selection may be derived from the VLF signal itself, for instance by applying a cryptographic hash function to the most recent VLF environmental fingerprint in conjunction with a publicly known service list.

The external time anchoring provides an upper time bound ("the cryptographic hash value existed no later than at this point in time"). The lower time bound is established by the device-local, offline-capable time references (in particular long-wave time signal, GNSS time, VLF environmental fingerprint). The recording device generates and signs provenance data locally, even without an existing network connection, and buffers these outside the first sector. When a network connection exists or is established, the buffered data are transmitted to the external services. The recording device is thus autonomous in its

core cryptographic operation; the external anchoring constitutes an additional hardening measure, not a prerequisite.

Further preferably, the recording device autonomously captures connection metadata and includes these in the hash value computation. These connection metadata are added to the data package in the second sector prior to the second signing and are hashed together. These metadata may comprise in particular the network address, TLS certificate chains of contacted external services, DNS resolution data, and/or results of a traceroute performed by the recording device itself. They offer forensic value, as they can be retroactively correlated with records of network operators, insofar as such records are available. The cooperation of network operators is, however, not a prerequisite of the inventive method, but an additional, optional validation possibility.

If an inventive camera has self-collected GNSS data, and its time is cross-validated by a time signal transmitter, and in addition a VLF environmental fingerprint is collected and co-signed, a highly reliable proof of provenance can be established entirely without time anchoring by external services, provided the VLF environmental fingerprint can be traced. Since a sufficiently dense and forensically reliable VLF reference infrastructure is not yet available on a comprehensive basis at present, the external time anchoring according to the preceding explanations constitutes for the time being the practically preferred method of external time binding, while the VLF environmental fingerprint serves as a complementary, offline-capable, and third-party-independent proof of provenance.

15. Recovery from Bricking (Bricking Recovery)

Insofar as a recording device according to the invention has tamper protection according to Section 4, which upon detected or suspected manipulation triggers an immediate zeroization of the private keys held in volatile memory—which is strongly recommended with regard to system security—a false triggering thereof leads to an unintended bricking, i.e., an irreversible loss of the cryptographic identity of the recording device. Such a false triggering of the tamper protection would inevitably occur also during ordinary repairs, for example when opening the housing for a battery replacement, a lens change, or other maintenance. Without further measures, an inventive camera would thus be a “disposable camera” whose cryptographic operation irretrievably ends after the first housing opening.

To remedy this, it is proposed to provide in both the first sector and the second sector a plurality of PUFs (per signature engine), for example five to ten discrete PUF structures per sector. During the initial initialization of the recording device, only a single PUF per signature engine is used for key generation according to Section 3.1. Its entropy contribution is combined as described with at least one further key component, whereupon the PUF hardware is irreversibly destroyed. The private key thus formed then exists exclusively in volatile memory of the respective sector.

If this volatile memory is zeroized—whether by intentional triggering of the tamper protection, by a repair procedure, or by a loss of supply voltage—the recording device can be re-initialized by using in each sector a next, hitherto unused PUF for the generation of a new key pair, which is again irreversibly destroyed after key generation. Each re-initialization thus generates an entirely new cryptographic device identity. The number of PUFs provided in the recording device per sector determines the maximum number of possible re-initializations.

In principle, it would also be conceivable that, instead of complete destruction of a discrete PUF, an irreversible physical alteration is made to a single PUF structure, which leads to a new entropy

contribution independent of the preceding one. This would be advantageous insofar as a single PUF structure could be used multiple times, or even practically an arbitrary number of times. However, ensuring the independence and entropy quality of successive entropy contributions from such a modified structure is likely to be comparatively much more difficult to guarantee than when using a fresh, untouched, discrete PUF. The sequential use of a plurality of discrete PUFs is therefore the preferred embodiment.

A side effect of this approach is that a repair that permits continued inventive cryptographic camera operation is, at least with respect to the first sector, technically mandatory to be performed only by the manufacturer: Only the manufacturer may issue the manufacturer-side key component according to Section 3.1 (this process can and should be cryptographically secured), and only the manufacturer can authenticate itself to the blockchain(s) in which the public keys of the recording device are published and the public keys of the first sector are cryptographically linked with those of the second sector (Section 7.1). Thus, from the sectored architecture and the PUF-based key management, a system-inherent manufacturer binding for repairs follows, which is a hard technical consequence of the security architecture. The manufacturer binding in the second sector is, by contrast, less mandatory: Since the second sector can be configured as an open-source system, and the user-side key component is assigned to the user, the manufacturer can enable the user to re-initialize the second sector with the cooperation of the blockchain infrastructure.

All recordings generated prior to a re-initialization remain fully verifiable, since the associated public keys remain permanently deposited in the blockchain. After re-initialization, the recording device merely receives a new cryptographic identity, the linkage of which with the preceding identity of the same device can likewise be deposited in the blockchain, in order to make the device history seamlessly traceable.

16. Implementation Notes on Side-Channel Suppression

16.1 Power Supply of the First Sector

A power supply of the first sector from the second sector is in part unavoidable, but constitutes a breach of the otherwise very strictly sectored security topology. This appears problematic in general, but particularly so in view of the proposed call-and-response communication during lifelong self-observation of the camera. To this end, it is first of all proposed to treat the matter as an EMC problem, and to provide at least one low-pass filter that is inaccessibly located within the first sector. Preferably, the following procedure is adopted: If the first sector has an electrically conductive housing, which is very strongly preferred, this housing is placed on a common reference ground of the first and second sectors, so that only a single conductor needs to be routed into the protected proprietary sector. Preferably, the metallic housing of the first sector comprises an additional chamber, so to speak a Faraday cage within the Faraday cage of the first sector. The conductor required for the power supply of the first sector can be routed into this chamber by means of a feedthrough capacitor and there charge a buffer capacitance that functions as a low-pass filter. This buffer capacitance shall be dimensioned by a person skilled in the art, have a low self-inductance, and is preferably operated in parallel with a TVS diode. This buffer capacitance is effective between the housing and the conductor from the second sector. Behind this buffer capacitance there is at least one regulator, namely at least a linear regulator, but preferably a switching converter with galvanic isolation between input and output, wherein the regulator shall operate autonomously, i.e., without a data connection to any other component or assembly, wherein the voltage regulation is

performed exclusively on the output side, and the switching converter is preferably operated at a fixed switching frequency. The output voltage of the regulator is in turn capacitively buffered, wherein this rear buffer capacitance is, preferably, again operated in parallel with a TVS diode, and is finally routed via a further feedthrough capacitor into the "main chamber" of the first sector, in which the signature engine, image sensor, etc., may be located. Additionally, inductors may be provided to form a low-pass filter together with the aforementioned buffer capacitances. If dimensioned sufficiently large, the rear buffer capacitance functionally replaces a dedicated power supply for the first sector: Transients introduced from a compromised second sector "disappear in the noise," i.e., they can no longer be correlated with bit operations.

16.2 Optical Data Diode

If the first sector has a metallically conductive housing, this is also advantageous with respect to an optical data diode according to claim 1: If its optical waveguide or light beam is routed through a bore in the metallic housing of the first sector into the second, parasitic component capacitances within the first sector are shielded from the second sector.

16.3 Pyrofuze Wire as a Potential Side Channel

A further potential side channel exists precisely when using a Pyrofuze fuse, since it can in principle also transmit an electrical signal. This can be remedied by grounding the Pyrofuze wire. If, as described in the preceding paragraph, the housing of the first sector is electrically conductive and is used as ground, it can simply be ensured that the Pyrofuze wire, when routed into the first sector, is in reliable electrically conductive contact with its housing, wherein the contact area between the Pyrofuze wire and the housing must be dimensioned such that the heat dissipation at the contact point does not inhibit the propagation of the exothermic reaction.

17. Radio Fingerprinting

17.1 VLF Fingerprinting

In the inventive VLF fingerprinting, it would be erroneous to assume that in the VLF band, for example between 3 kHz and 30 kHz, exclusively signals of natural origin would be observable. On the contrary, in densely populated areas, the most intense signal sources in this band may be of anthropogenic origin, for example caused by nearby switching power supplies, frequency converters, or power electronic installations; these signals, too, can be used for provenance verification. However, the method also works without anthropogenic VLF sources, since natural sources are always and everywhere present.

In the following, a preferred embodiment of a VLF receiver is described, which is intended to enable a person skilled in the art to carry out the invention, without limiting the claimed teaching to the specifically described configuration. In particular, modifications familiar to a person skilled in the art, substitutions of individual components, or alternative dimensioning may be made without departing from the scope of protection of the invention.

A VLF receiver suitable for a mobile terminal device preferably comprises a soft-magnetic rod antenna with a receiver coil. The rod antenna can be realized in a compact form factor that can be integrated into common portable devices, since for the spectral fingerprint sought herein no high absolute sensitivity is required, but merely a sufficient distinguishability of the signal sources contained in the spectrogram. As a

core, a soft-magnetic ferrite rod is particularly suitable, for example one made of MnZn ferrite. Instead of ferrite, laminated or powdered cores of highly permeable nanocrystalline or amorphous soft-magnetic alloys may also be used, for example materials such as Nanoperm (Vitroperm) or Metglas. In an open magnetic circuit, as presented by a simple rod core, the effective permeability μ_{eff} is, however, limited by the demagnetization factor N: For $\mu_r \gg 1/N$, μ_{eff} converges toward the value 1/N and is then nearly independent of the material permeability μ_r. Since at practically common rod length-to-diameter ratios the effective permeability already saturates at moderate material permeabilities, highly permeable core materials in open rod geometry offer no substantial reception advantage over MnZn ferrites with μ_r in the range of several thousand. Nevertheless, they may be expedient due to their substantially lower Barkhausen noise compared to ferrites, since the much smaller hysteresis of these materials compared to ferrites suppresses the Barkhausen noise in the core material caused by abrupt domain wall displacements, which in a receiver designed for low signal levels can improve the achievable noise floor. Furthermore, these materials exhibit a higher saturation magnetization and a lower magnetostriction (and correspondingly lower microphonics).

For the receiver coil, a low-capacitance winding technique is preferably employed, in particular a section winding (chamber winding), in which the winding is subdivided into axially separated sections. A section winding executed by a person skilled in the art substantially reduces the parasitic winding capacitance compared to a continuous layer winding, because the potential of adjacent turns within a section differs only slightly, while the section walls reduce the capacitive coupling between sections of different potential. The receiver coil is operated below its self-resonant frequency as a pure inductance. That is, the inductive reactance of the coil exceeds in magnitude at all operating frequencies the capacitive impedance of the parasitic winding capacitance. The number of turns is therefore preferably chosen high enough that the self-resonant frequency of the coil still lies above twice the highest design receiving frequency. This leaves a sufficient margin to the upper band edge, so that the transfer function in the useful band remains largely frequency-independent. If the receiver coil is arranged on the same rod core as a time signal receiver coil that is tuned to a time signal frequency, for example 77.5 kHz for DCF77, the self-resonant frequency of the VLF receiver coil shall furthermore be sufficiently spaced from this time signal frequency and its harmonics, in order to avoid an undesired resonance peaking and thus an overdriving of the subsequent amplifier stage.

If the arrangement is housed in a device that contains further electronic assemblies, electromagnetic shielding of the rod antenna and the preamplifier against device-internal interference sources, in particular clock oscillators and switching regulators, may be expedient.

The signal from the receiver coil is preferably fed to a low-noise differential amplifier, in particular one with low 1/f noise, for example an operational amplifier with a JFET input stage. Both terminals of the receiver coil are connected to the two inputs of the differential amplifier; the coil thereby serves simultaneously as the signal-delivering element and as a balanced source. A common-mode signal, such as an electrostatic coupling, produces at both coil terminals approximately the same potential and is attenuated by the common-mode rejection of the differential amplifier, while the magnetically induced useful signal appears as a differential voltage between the terminals. The gain-bandwidth product of the differential amplifier is preferably dimensioned such that at maximum signal gain a sufficient headroom remains, in order to avoid clipping of the output signal even at sporadically elevated field strengths, for example from nearby interference sources.

Optionally, the inputs of the differential amplifier may be protected against overvoltages by TVS diodes of low capacitance, for example against transients such as those that may be induced by nearby lightning discharges. A low capacitance is herein understood to mean one that is lower than, and preferably several times lower than, the stray capacitance of the receiver coil, so that the TVS diodes do not substantially affect the resonant frequency of the coil and the transfer function of the receiver.

The output signal of the differential amplifier is preferably band-limited by a low-pass filter in order to suppress signal components above the VLF useful band and to reduce the dynamic range requirements on the subsequent analog-to-digital converter. For avoidance of aliasing, in particular signal components with frequencies above half the sampling rate of a subsequent ADC are to be suppressed. The band-limiting may be effected by means of an active low-pass filter in multiple-feedback topology (MFB filter), wherein the MFB filter preferably has an adjustable voltage gain for level matching.

The thus obtained, optionally filtered and level-matched signal is digitized by means of an analog-to-digital converter. The digitized signal is subjected to a short-time Fourier transform (STFT), wherein the temporal window length is chosen such that the spectral resolution in the VLF band is sufficient for discriminating the present signal sources, and at the same time the temporal resolution enables an association with a specific recording time. The resulting time-dependent spectrogram constitutes the sought VLF fingerprint and is co-signed in the second sector.

17.2 Shortwave Fingerprinting

An inventive radio fingerprinting is not only possible in the VLF band. A most interesting alternative is to be seen in higher-frequency anthropogenic signals, and in particular in military, maritime, and intelligence shortwave transmitters, including numbers stations, and further services such as VOLMET aviation weather reports transmitted on shortwave. This is interesting not only because it is practically impossible that these very diverse sources with partly diverging interests would collude to produce a specific signal at a specific location: They could not do so, at least not completely, because shortwave transmission depends on unpredictable variables with weather and solar activity: Shortwave signals reach the receiver via multiple propagation paths simultaneously, namely as a ground wave and as singly and multiply ionospherically reflected “sky waves,” optionally via different ionospheric layers (E, F1, F2). These partial signals travel paths of different lengths and therefore arrive at the receiver with different phase relationships. Since the ionosphere is not a static reflector but its layer heights and electron densities vary continuously with time of day, solar activity, and atmospheric dynamics (weather), the relative phase relationships of the partial signals change continuously. The resulting interference alternates between constructive and destructive superposition, which is observed at the reception site as temporally and spectrally varying signal strength fluctuations, known as a “fading pattern.” In addition, intense and non-predictable natural and anthropogenic interference is present in the shortwave band as well.

For collecting a shortwave fingerprint, for example, a software-defined radio (SDR) is suitable, which by means of a broadband active antenna observes a plurality of publicly known shortwave transmitters, wherein the signal of each transmitter is, e.g., digitally demodulated and subsequently subjected to a Fourier transform (STFT). Together with the temporally varying relative signal strengths, the determined time-dependent spectra form the fingerprint or its basis. Such fingerprinting is suitable both for recording devices according to claim 1 and for an illumination device according to claim 62. The described shortwave fingerprinting, on the transmitter side, makes use of existing, predominantly governmental, strictly classified, and highly available infrastructure, without subjecting itself to the truthfulness of the

messages transmitted by these sources; rather, these existing transmitters can be used to probe the ionosphere, and this with commercially available components.

18. GNSS Protocol Extension

In an advantageous further development, it is proposed that navigation satellites in the future continuously co-transmit, as part of their civilian services, true random numbers or cryptographically strong, unpredictable pseudo-random numbers, wherein preferably each satellite transmits its own random sequence. These random numbers are received by a multitude of independent receivers worldwide and can be publicly observed, archived, and indexed by the operators of the navigation systems and/or independent third parties, so that at any point in time it can be reconstructed which random sequence was transmitted by which satellites at that time.

For authenticity verification, this provides a significant advantage: Since the random numbers are by construction unpredictable, and their repetition, given a sufficiently long length of the random sequences, appears to be practically excluded under realistic conditions, a replay attack—in which manipulated GNSS signals with previously transmitted random sequences are fed to a recording device—would be detected upon subsequent verification by the fact that the same random sequence had already been observed and indexed in the past. Any repetition of an archived random sequence at a later point in time can thus be interpreted as an indication of a signal replay.

A recording device according to the invention is configured to sign the navigation data received by its GNSS receiver in the second sector, including the respectively valid random numbers, as part of the metadata, together with the recording data, further metadata of the second sector, and the first signature and optionally metadata of the first sector, with the private key of the second sector. Thereby, the recording is cryptographically bound to a concretely observed state of the GNSS system:

The resulting, GNSS-side random-number-based metadata can be advantageously combined with external provenance protocols, in particular protocols according to the Content Provenance Profile (CPP) and/or C2PA: While the provenance protocol provides an external, publicly verifiable time anchoring and certificate chain, the GNSS random numbers signed in the camera provide an additional, physically unpredictable state anchor of the satellite system, so that replay-based GNSS spoofing attacks can be detected with high confidence.

Claims

Group A: Sectored Architecture and Data Diode (Claims 1–13)

Independent Claim 1 – Sectored Recording Device

1. A recording device for generating tamper-resistant data, comprising:

a) a first sector comprising at least one sensor and a first signature engine, the first signature engine being configured to determine cryptographic hash values from a signal captured by the sensor, and to digitally sign said hash values with a private key of a key pair assigned to the first sector, so as to generate a first signature;

b) a second sector comprising at least a second signature engine, the second signature engine being configured to receive the signal and the first signature and to generate a second signature using a private key of a key pair assigned to the second sector;

c) a data diode disposed between the first sector and the second sector, which effects a physically enforced unidirectional data transmission from the first sector to the second sector and prevents information backflow from the second sector to the first sector;

wherein the first sector and the second sector are assigned to different trust domains.

2. The recording device according to claim 1, wherein the first sector is assigned to the manufacturer of the recording device and the second sector is assigned to the user of the recording device.

3. The recording device according to any one of the preceding claims, wherein the sensor comprises an image sensor and the signal is an audiovisual signal.

4. The recording device according to any one of the preceding claims, wherein the data diode is realized by at least one of:

a) an optoelectronic coupler, in which a light-emitting component in the first sector drives a photosensitive component in the second sector;

b) a fiber-optic connection with exclusively one light-transmitting unit and one light-receiving unit;

c) a circuit with strongly asymmetric transfer characteristics, which physically prevents signal transmission in the reverse direction or attenuates it to such an extent that information backflow is prevented.

5. The recording device according to any one of the preceding claims, wherein the first sector is arranged wholly or partially within the second sector.

6. The recording device according to any one of the preceding claims, wherein the first sector is configured in a manufacturer-specific and proprietary manner and the second sector is configured as an open-source system whose hardware and/or software specification is publicly viewable and verifiable.

7. The recording device according to any one of the preceding claims, wherein the first signature engine is configured to include in the hash value computation and to sign, jointly with the sensor data, metadata

generated in the first sector—in particular autofocus information, depth sensor data, and/or liveness parameters—before said data leave the first sector.

8. The recording device according to any one of the preceding claims, wherein the first signature engine is integrated as a system-on-chip (SoC) on the chip of the image sensor, so that the signal path between data source and signing point is minimized.

9. The recording device according to any one of the preceding claims, wherein in the first sector, a RAW data stream and at least one compressed data stream are generated in parallel and both data streams are independently signed before being communicated to the second sector via the data diode.

10. The recording device according to any one of the preceding claims, wherein the second sector further comprises a plurality of sensors selected from: a GNSS module, a long-wave time signal receiver, inertial sensors including gyroscopic sensors and accelerometers, environmental sensors, and wherein the metadata captured by these sensors are added to the signal prior to the second signing.

11. The recording device according to any one of the preceding claims, wherein the recording device is configured to activate and deactivate the first sector by controlling the supply voltage from the second sector, it being ensured by hardware means that no changes to a firmware of the first sector can thereby be effected.

12. The recording device according to claim 11, wherein the firmware of the first sector distinguishes between at least two operating modes based on the duration of the applied supply voltage:

a) a challenge-response mode, in which upon a voltage pulse having a duration below a predetermined threshold duration, a single signed image is generated and output via the data diode; and

b) a regular recording mode, in which upon a supply voltage applied beyond the threshold duration, continuous recording operation is commenced.

13. The recording device according to claim 12, wherein the second sector monitors the response time of the first sector to a voltage pulse and evaluates a failure of the expected response to arrive or an impermissible deviation of the response time from a time profile characteristic of the first sector as a manipulation indicator.

Group B: Key Generation and PUF Destruction (Claims 14–18)

Independent Claim 14 – PUF-Based Key Generation with Irreversible PUF Destruction

14. A method for key generation for a cryptographic system, comprising the steps of:

a) obtaining a hardware-bound entropy contribution by means of a Physical Unclonable Function (PUF) that exploits manufacturing-related variations of a semiconductor structure;

b) combining the entropy contribution with at least one further key component to generate a private key;

c) storing the private key in a volatile memory;

d) after completion of the key generation, irreversibly destroying the PUF hardware so that a renewed obtaining of the entropy contribution is physically impossible;

wherein after step d) the private key exists exclusively in the volatile memory.

15. The method according to claim 14, wherein the irreversible destruction of the PUF hardware is effected by applying a voltage to at least one MOS device of the PUF that produces an irreversible gate oxide breakdown.

16. The method according to claim 14 or 15, wherein the private key is composed of at least three components: the PUF entropy contribution, a piece of partial information introduced by the manufacturer, and a piece of partial information introduced by the user, so that neither the manufacturer, nor the user, nor the PUF hardware alone knows or can reconstruct the complete private key.

17. The method according to any one of claims 14 to 16, wherein the volatile memory is an SRAM memory that is permanently powered by a battery, and wherein a loss of supply voltage leads to the immediate and irreversible loss of the private key.

18. The recording device according to any one of claims 1 to 13, wherein in at least one of the sectors a private key is generated and stored according to the method of any one of claims 14 to 17, and wherein preferably in each of the sectors a respective private key is generated and the respective PUF is irreversibly destroyed, so that after initialization the private keys exist exclusively in volatile memories in at least two disjoint sectors.

Group C: Gas Pressure Monitoring (Claims 19–23)

Independent Claim 19 – Gas Pressure Monitoring for Tamper Detection

19. An electronic device with tamper protection, comprising:

a) at least one memory in which security-relevant data are held;

b) at least one gas-tight housing in which the memory is arranged and which is filled with a gas whose pressure deviates from an ambient pressure;

c) at least one pressure sensor and at least one temperature sensor, which measure the pressure and the temperature of the gas quasi-continuously;

d) an evaluation unit configured to trigger a zeroization of the security-relevant data as soon as at least one of the following conditions is detected:

i) a rate of pressure change that exceeds a predetermined threshold value;

ii) an absolute pressure outside a permissible pressure range;

iii) a deviation between the measured pressure and a pressure expected on the basis of the measured temperature and a known volume according to the ideal gas law that exceeds a predetermined tolerance range;

iv) a temperature below a predetermined minimum value, in particular for protection against cold-boot attacks.

20. The electronic device according to claim 19, wherein the gas is an inert gas and/or wherein the chemical composition of the gas fill is monitored by means of electrochemical or optical sensors, the exact composition being a manufacturer secret.

21. The electronic device according to claim 19 or 20, wherein the evaluation unit is further configured to initiate, upon detection of a manipulation, in addition to data zeroization, a thermal destruction of the memory, in particular by short-circuiting a battery powering the memory.

22. The electronic device according to any one of claims 19 to 21, wherein upon loss of housing integrity, a chemical destruction of the memory is provided by reaction of memory components or a reactive substance surrounding them with atmospheric constituents.

23. The recording device according to any one of claims 1 to 13 and 18, wherein at least one sector is configured as an electronic device according to any one of claims 19 to 22.

Group D: Cross-Sector Key Zeroization (Claims 24–27)

Independent Claim 24 – Key Zeroization by Diffusion-Controlled Chemical Reaction

24. An electronic device having at least two spatially separated security domains, wherein:

a) in each security domain, at least one volatile memory is arranged that contains security-relevant data and is powered by an associated battery, wherein each battery has an associated fuse;

b) the fuses of the different security domains are thermally coupled by at least one contiguous reactive element, the reactive element having a first portion in a first security domain and a second portion in a second security domain;

c) a triggering of the fuse in one security domain initiates an exothermic, self-sustaining chemical reaction of the reactive element, which propagates along the reactive element and inevitably co-triggers the fuse of the other security domain;

so that a command for key zeroization crosses a security domain boundary as a diffusion-controlled chemical reaction, without an electrical or electromagnetic signal being transmitted.

25. The electronic device according to claim 24, wherein the reactive element comprises a wire of an aluminum-palladium alloy (Pyrofuze) or a reactive multilayer sheet based on aluminum and nickel (NanoFoil).

26. The electronic device according to claim 24 or 25, wherein the reactive element additionally serves as a fuse for one or more of the batteries.

27. The recording device according to any one of claims 1 to 13, 18, and 23, wherein the first and the second sector are thermally coupled by a reactive element according to any one of claims 24 to 26.

Group E: Permanent Cryptographic Self-Observation (Claims 28–32)

Independent Claim 28 – Cryptographic Image Chain

28. The recording device according to any one of claims 1 to 13, wherein the recording device is configured to generate, throughout its entire operating life—including outside of regular recording operation—at time intervals whose expected value preferably lies in the range of seconds to a few minutes, compressed background images by means of the sensor of the first sector, wherein:

a) each background image is signed close to the source in the first sector, unidirectionally communicated to the second sector via the data diode, and there subjected to a second signing together with current metadata;

b) each chain element is cryptographically chained with the preceding one by including the cryptographic hash value of the preceding chain element in the subsequent one;

c) the resulting image chain is stored locally in the second sector in an encrypted manner;

so that a seamless, tamper-evident image chain is created over the entire operating life of the recording device.

29. The recording device according to claim 28, wherein the time intervals between the background images are determined pseudo-randomly.

30. The recording device according to claim 28 or 29, wherein a Merkle tree is formed over the chain elements, which enables a verifiable disclosure of a temporal section of the image chain without having to disclose the entire chain.

31. The recording device according to any one of claims 28 to 30, wherein the image chain further comprises the challenge-response frames generated according to claim 12.

32. The recording device according to any one of claims 28 to 31, wherein the image chain is stored in a symmetrically encrypted manner, the symmetric key being known exclusively to the user.

Group F: Anti-Screen-Recording and Liveness (Claims 33–34)

33. The recording device according to any one of claims 1 to 13, wherein the first sector further comprises at least one of the following systems for detection of screen reproductions:

a) a depth sensor, wherein preferably an existing autofocus system is incorporated;

b) a system with active object illumination, in particular in the near-infrared range;

c) an ultrasound interaction system for determining three-dimensional object properties;

d) a liveness sensor, in particular an image sensor configured as an RGB-IR sensor (4-channel sensor), which simultaneously captures visible light and infrared radiation;

wherein signals from these sensors are signed close to the source in the first sector.

34. The recording device according to claim 33, wherein imaging sensors are assigned to the first sector and illumination devices comprising a light source are assigned to the second sector, so that a compromise of the second sector does not impair the authenticity of the imaging sensors.

Group G: Electromagnetic and Chemical Protection (Claims 35–38)

Independent Claim 35 – Electromagnetic and Chemical Protection of a Cryptographic Assembly

35. An electronic assembly with cryptographic function, comprising:

a) a cryptographic circuit embedded in a potting compound, the potting compound being selected such that:

i) it is insoluble in organic and halogenated solvents at standard conditions; and

ii) all solvents, including supercritical fluids, that can substantially swell or dissolve the potting compound also attack and functionally destroy the potted electronics;

b) an electromagnetic shield at least partially surrounding the cryptographic circuit, wherein for optical feedthroughs an electrically conductive, optically transparent coating, in particular an ITO coating, is provided;

c) a consistently differential signal transmission arranged within the shield, at least outside of integrated circuits, for suppression of common-mode interference.

36. The electronic assembly according to claim 35, wherein the potting compound contains fillers for influencing its thermal conductivity and/or electromagnetic properties, in particular aluminum nitride (AlN) and/or ferrite powder.

37. The electronic assembly according to claim 35 or 36, wherein the electrically conductive, optically transparent coating is applied on the rear lens element of an objective lens and, in conjunction with a conductive housing, forms a substantially closed Faraday cage, the coating being connected to the housing via a permanent seal.

38. The recording device according to any one of claims 1 to 13, wherein at least the first sector is configured as an electronic assembly according to any one of claims 35 to 37.

Group H: Data Diode Protection and Signal Path Monitoring (Claims 39–42)

39. The recording device according to any one of claims 1 to 13, wherein at least one tamper protection device of one of the sectors is configured to also detect manipulation attempts that affect a transmission path of the data diode that lies outside both sectors.

40. The recording device according to any one of claims 1 to 13, wherein at least in the second sector a key zeroization is triggered as soon as an expected signal from the data diode fails to arrive, the expected signals preferably following one another at pseudo-random intervals.

41. The recording device according to any one of claims 1 to 13, wherein at least one signal path between the sensor and the first signature engine within the first sector is monitored and a change of this signal path is evaluated as a manipulation indicator.

42. The recording device according to any one of claims 1 to 13, wherein upon detection of an electromagnetic attack, in particular an overvoltage event, an immediate key zeroization is triggered.

Group I: PQC and Blockchain (Claims 43–45)

43. The recording device according to any one of claims 1 to 13, wherein at least one of the signature engines uses a post-quantum cryptography algorithm (PQC algorithm).

44. The recording device according to claim 43, wherein the first and the second signature engine use different PQC algorithms that are based on different mathematical problems, in particular selected from: lattice-based problems, code-based problems, hash-based schemes, and/or multivariate polynomial equation systems.

45. The recording device according to any one of claims 1 to 13, wherein the public keys of the first and the second sector are deposited in a linked manner in a plurality of mutually independent blockchains, so that an attacker would have to simultaneously overcome the independent consensus mechanisms of a plurality of blockchains.

Group J: Atmospheric Signals (Claims 46–48)

Independent Claim 46 – Authenticity Verification by Means of Atmospheric Electromagnetic Signals

46. A system for authenticity verification of a data set, comprising:

a) at least one sensor for capturing an electromagnetic environmental signal in a frequency range that is characteristic of atmospheric and/or ionospheric propagation conditions, in particular in the VLF range (Very Low Frequency) or in the shortwave range;

b) a processing unit configured to:

i) determine from the captured electromagnetic environmental signal at least one characteristic parameter or a feature pattern, in particular characteristic quantities of sferics, tweeks, and/or VLF hiss, and/or spectral power densities, and/or time-dependent spectrograms; and

ii) associate the parameter or the feature pattern with a recording event temporally and spatially;

c) a comparison unit configured to:

i) provide reference data relating to electromagnetic environmental signals, the reference data having been captured and aggregated by at least one reference receiver and/or by an external observation infrastructure; and

ii) compare the determined parameter or the feature pattern with the reference data and to derive from the comparison a measure for the plausibility of the asserted location and/or time of the recording event;

wherein the system is configured to make, on the basis of the measure, a statement about the authenticity of the data set with respect to its location and/or time indication.

47. The system according to claim 46, wherein the reference receiver is a further recording device, in particular a recording device according to any one of claims 1 to 13.

48. The system according to claim 46 or 47, wherein the external observation infrastructure comprises at least one of an Earth observation satellite, a network of geophysical sensors, or a lightning location network.

Group K: Platform System for Authenticity Verification (Claims 49–50)

Independent Claim 49 – Platform System

49. A data processing system, in particular a video platform, social media platform, web browser, or browser application, configured to:

a) verify, for a received audiovisual data element, whether it can be attributed through an unbroken certificate chain to a recording made with a recording device that has a multi-stage signature chain based on at least two independent trust domains;

b) signal to a user whether the audiovisual data element has been verified as authentic; and/or

c) enable the user to activate a filter in which preferably only recordings verified as authentic are displayed.

50. The data processing system according to claim 49, wherein the verification according to feature a) comprises checking a first signature of a proprietary sector and a second signature of an open sector against cryptographically linked public keys deposited in at least one blockchain, and wherein the verification of both signatures is required for a positive authenticity assessment.

51. A method for external time anchoring of recording data of a recording device, comprising the steps of:

a) generating cryptographic hash values from a signal captured by the recording device;

b) transmitting the cryptographic hash values to a plurality of mutually independent device-external time-anchoring services;

c) receiving time-anchoring receipts from the services;

wherein the selection of the specific services from a service list is made by a procedure whose result is retroactively verifiable but not predictable in advance.

52. The method according to claim 51, wherein the plurality of services belong to different categories, in particular Time Stamping Authorities (TSA), publicly auditable Transparency Logs, and blockchains.

53. The method according to claim 51 or 52, wherein the selection of the specific services is derived by applying a cryptographic hash function to an unpredictable environmental signal captured by the recording device, in particular an atmospheric electromagnetic signal in the VLF range, in conjunction with a publicly known service list.

54. The method according to any one of claims 51 to 53, wherein the recording device generates and buffers the cryptographic hash values and associated provenance data locally, and the transmission to the external services occurs only when a network connection exists or is established, wherein device-local time references establish a lower time bound and the external time anchoring establishes an upper time bound for the time of formation of the hash values.

55. The method according to any one of claims 51 to 54, wherein the recording device autonomously captures connection metadata of the communication with the external services, in particular its own network address, TLS certificate chains of contacted services, DNS resolution data, and/or results of a traceroute performed by the recording device itself, and includes these connection metadata in the cryptographic hash value computation.

56. The recording device according to any one of claims 1 to 13, which is further configured to carry out a method according to any one of claims 51 to 55, wherein the transmission to the external services and the processing of received time-anchoring receipts occur outside the first sector.

57. A cryptographic system, comprising:

a) N Physical Unclonable Functions (PUFs), where $N \geq 2$;

b) a volatile key memory;

c) a control unit configured to, upon each initialization or re-initialization of the system:

i) select a next, hitherto unused PUF;

ii) obtain an entropy contribution from the selected PUF and combine it with at least one further key component to form a private key;

iii) store the private key in the volatile key memory; and

iv) irreversibly destroy or irreversibly physically alter the selected PUF;

wherein N determines the maximum number of possible initializations and each initialization generates an identity that is cryptographically independent of all preceding initializations.

58. The cryptographic system according to claim 57, wherein upon loss of the contents of the volatile key memory, in particular by triggering of a tamper protection device, by a repair procedure, or by a loss of supply voltage, a re-initialization according to feature c) is performable with a next unused PUF.

59. The cryptographic system according to claim 57 or 58, further comprising an interface to at least one blockchain, wherein:

a) upon each initialization, a public key associated with the private key is deposited in the blockchain; and

b) upon a re-initialization, the new cryptographic identity is linked with the preceding identity of the same system in the blockchain;

so that all signatures generated prior to a re-initialization remain verifiable by means of the public keys permanently deposited in the blockchain and the device history is seamlessly traceable.

60. The recording device according to any one of claims 1 to 13 and 18, wherein in at least one sector a cryptographic system according to any one of claims 57 to 59 is arranged, wherein preferably in each sector a plurality of PUFs per signature engine is provided, and wherein a re-initialization of the first sector requires the introduction of a manufacturer-side key component.

61. The recording device according to claim 34, wherein at least one illumination device comprising a light source of the second sector is configured to illuminate the object to be recorded with modulated light carrying data provided by the second sector, in particular geolocation data, time information, and/or an environmental fingerprint, in particular a radio environmental fingerprint, and in particular a VLF fingerprint, wherein the image sensor of the first sector captures the modulated illumination as part of the recorded signal, which is subsequently signed in the first sector.

62. An illumination device comprising a light source for data-carrying object illumination, configured for use with a recording device having cryptographic signing, wherein the illumination device can illuminate an object to be recorded with modulated light carrying provenance data, in particular geolocation data, time information, and/or an environmental fingerprint, in particular a radio environmental fingerprint, and in particular a VLF fingerprint.

63. A recording device for generating tamper-resistant recording data, comprising:

a) at least one sensor for capturing image and/or audio data;

b) a GNSS receiver configured to receive navigation data of a satellite navigation system, the navigation data comprising at least one sequence of true random numbers or cryptographically strong, unpredictable pseudo-random numbers provided by the satellite navigation system;

c) a signature engine configured to:

i) form a data set package from the recording data captured by the sensor and metadata,

ii) include the navigation data received by the GNSS receiver, including the respectively valid random numbers, as further metadata in the data set package, and

iii) form cryptographic hash values over the data set package and sign them with a private key assigned to the recording device,

wherein the random numbers contained in the data set package serve to cryptographically bind the signed data set to a concretely observed state of the satellite navigation system, so that upon subsequent authenticity verification by comparing the random numbers contained in the data set with random sequences archived by independent reference receivers, replay-based GNSS spoofing attacks through repetition of previously transmitted random sequences are detectable.

64. A method for registration of a digital recording device, comprising:

a) determining a digital fingerprint of an image sensor of the recording device, the digital fingerprint comprising a mapping of pixel defects and/or a signature derived from the Photo-Response Non-Uniformity (PRNU) of the image sensor;

b) associating the digital fingerprint with a public key of a cryptographic key pair assigned to the recording device; and

c) publishing the public key together with the associated digital fingerprint in at least one publicly auditable storage, in particular a blockchain,

so that upon subsequent authenticity verification of a recording made with the recording device, both the cryptographic signature of the recording is verifiable by means of the public key and the image sensor is forensically identifiable by means of the published digital fingerprint.

65. The recording device according to any one of claims 1 to 13, wherein upon registration of the recording device, a digital fingerprint of the image sensor of the first sector is determined, comprising a mapping of pixel defects and/or a signature derived from the Photo-Response Non-Uniformity (PRNU) of the image sensor, and wherein this digital fingerprint is associated with the public key of the first sector deposited in a blockchain and is published together therewith.